

Difficulty Analysis of Posiform-Planted QUBO Benchmarks via Energy and Hamming Metrics

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1 Motivation: the 0%-success wall

Benchmarking QUBO solvers needs instances whose optimum is *known* yet that stay *hard*. **Posiform planting** gives a unique planted optimum x^* ; a hardened variant adds random discrete-coefficient spin-glass QUBOs R_i and scales the planted posiform P by a coefficient α :

$$Q_\alpha(x) = \sum_i R_i(x) + \alpha P(x)$$

- Hardness is graded by the **ground-state success rate**.
- But across the whole hard regime that rate is **identically 0%**.
- ⇒ It **cannot rank** instances, solvers, or α *within* the hard regime — the benchmark loses its purpose.

2 Idea: measure the failures

Instead of “did it succeed?”, ask *how close the failing runs come to x^** , using two quantities — over **failures only**, on a **fixed set of the 100 hardest** instances:

- **Energy gap** $\Delta E = E(x) - E(x^*) \geq 0$ — depth of the trap.
- **Hamming distance** $\text{HD}(x, x^*)$ — bit-distance to the answer.

These turn the flat 0% into a **continuous, rankable difficulty curve**. Neither quantity is new — the **contribution is the protocol**: failure-conditioning on a fixed hardest set, validated on real hardware.

3 Methodology

Instances. 500 posiform-planted QUBOs on the D-Wave **Pegasus P16** graph ($n=5612$ qubits), lin2 coefficients $R_{ij} \in \{-1, +1\}$. Kernighan–Lin bisection into ≤ 16 -qubit blocks, each R_i solved exactly; $x^* = \text{concatenated block ground states (global min of } \sum_i R_i)$; P = posiform of a planted 2-SAT with unique solution x^* .

Fixed set. Rank instances by total SA failing-reads across the sweep; fix the hardest 100 — **chosen from SA alone, before any QPU run**, so the cross-solver agreement is *not circular*.

Metrics. $\text{lin2} \Rightarrow$ integer R energies $\Rightarrow$ report ΔE in units of the minimal R -gap $\Delta_R=1$ (comparable across instances). All statistics are **conditional on failure** ($x \neq x^*$).

Protocol. neal SA, 1000 sweeps, 100 reads / (instance, α). Cross-check: same 100 instances on a **D-Wave Advantage** annealer — hardware-native, auto-scale, two-gauge, 100 reads $\times 20 \mu\text{s}$.

4 Why report *both* metrics

At $\alpha=0$ (R alone, degenerate):

- **56.1%** of samples reach an R ground state ($\Delta E=0$)...
- ...yet **0%** equal x^* , and HD is *maximal* ($459 = 8.2\%$ of n).
- ⇒ ΔE alone reads “**solved**”; only **HD** reveals that no sample is near x^* .

As α grows:

- HD **decreases** (failures move *closer in bits* to x^*)...
- ...while ΔE rises to a **peak** (failures get *deeper in energy*).
- ⇒ **Opposite trends on the same samples** — each metric captures what the other misses.

5 The Δ_R floor and the αP peak

Failure-conditioned ΔE traces $\Delta E = \Delta R + \alpha \Delta P$ directly:

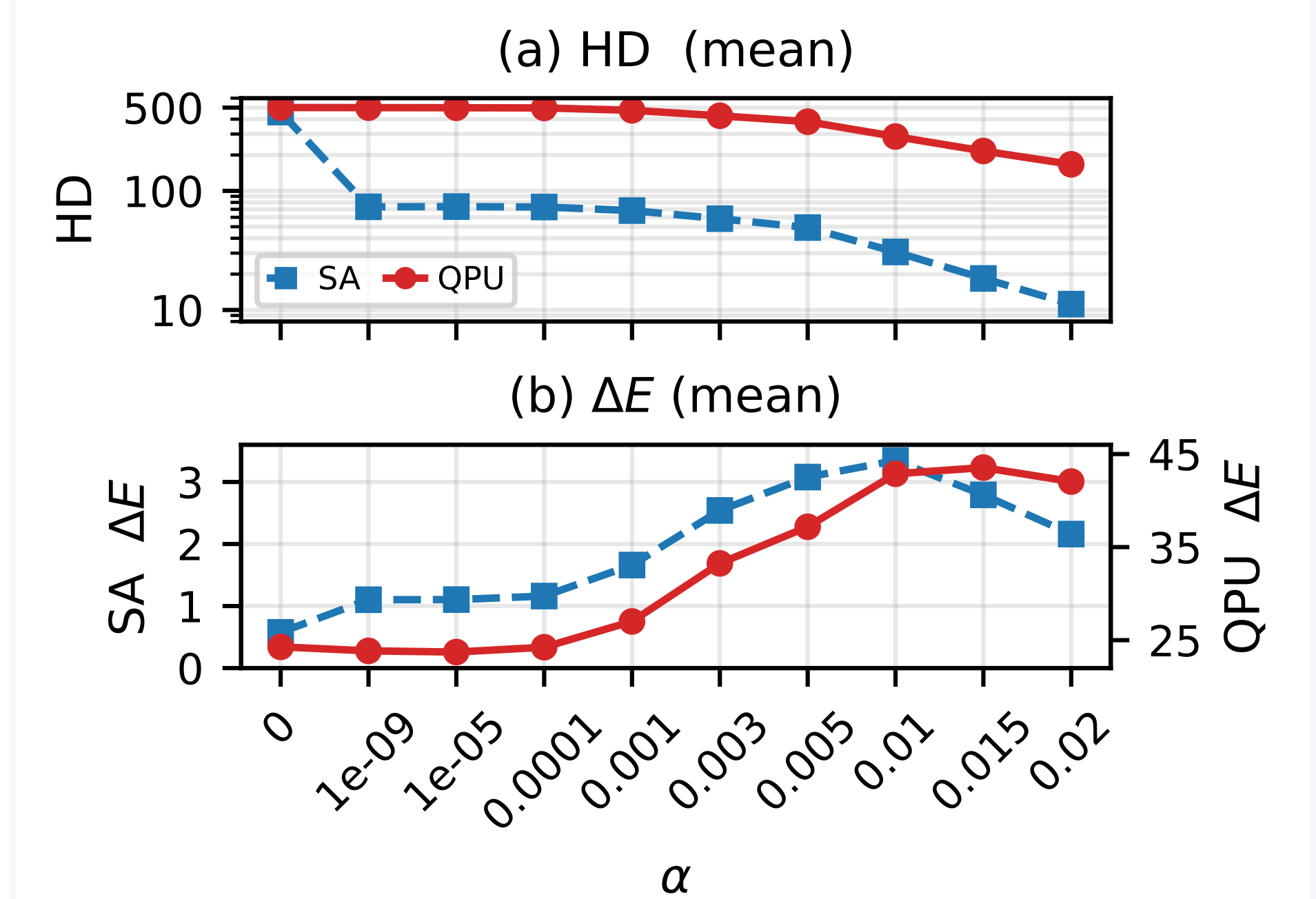
- **Floor.** For $\alpha \in [10^{-9}, 10^{-4}]$, ΔE is flat at the bare R -trap depth — mean 1.10, median exactly 1 (one Δ_R); $\alpha \Delta P$ negligible. HD flat ≈ 73 .
- **Peak.** Past $\alpha=10^{-3}$ the planted term switches on; ΔE rises to a **non-monotone peak of 3.35 at $\alpha=10^{-2}$** ($3.0 \times$ the floor), then declines to 2.16.
- **Mechanism.** $\Delta R, \Delta P$ are fixed per bitstring; as failures move toward x^* their ΔP falls, so ΔE peaks where rising α stops outpacing that fall.

Not a survivorship or selection artifact: the 100-instance set keeps failing at every α ($\geq 91\%$ of reads in aggregate), and the *same* non-monotone shape appears over **all 500 instances** (floor ≈ 1.1 , peak ≈ 3.0 , then 1.9) — the hardest set only sharpens it.

6 ΔE ranks what the success rate cannot

Two SA budgets (1000 vs. 5000 sweeps) both reach 0% success in the small- α region, yet mean ΔE **separates them** — at $\alpha=10^{-3}$: 1.68 **vs.** 1.00. The success rate is blind here; ΔE is not.

7 Fig. 1 — Failure-conditioned difficulty



Same 100 hardest instances (each α : $\geq 91\%$ of reads fail in aggregate). **Blue:** SA; **red:** D-Wave Advantage. **(a)** HD (log axis) decreases monotonically yet never reaches 0. **(b)** $\Delta E = \Delta R + \alpha \Delta P$ (split axes, left SA / right QPU): a flat Δ_R floor, then a non-monotone peak near $\alpha=10^{-2}$. The annealer reproduces the same qualitative shape on an order-of-magnitude larger baseline.

8 Table 1 — Wrong-only means (ΔE in Δ_R units)

α	HD		ΔE	
	SA	QPU	SA	QPU
0	458.8	501.0	0.58	24.3
10^{-9}	73.5	499.8	1.10	23.9
10^{-5}	73.8	498.8	1.10	23.7
10^{-4}	73.2	497.0	1.16	24.2
10^{-3}	68.3	474.5	1.66	27.0
3×10^{-3}	58.4	427.2	2.54	33.3
5×10^{-3}	49.2	380.9	3.08	37.2
10^{-2}	30.7	285.9	3.35	42.9
1.5×10^{-2}	18.4	216.1	2.79	43.5
2×10^{-2}	11.2	167.1	2.16	42.1

Peak ΔE in **orange**. SA: aggregate failure $\geq 91\%$ at every α ; QPU: 0% success at every α .

9 Quantum-annealer validation

Re-running the same 100 instances on **D-Wave Advantage** gives 0% success at every α , yet the **same qualitative pattern**: a non-monotone ΔE peak near $\alpha=10^{-2}$ and HD decreasing monotonically (**501 $\rightarrow$ 167**).

- Baselines an order of magnitude above SA ($\Delta E \approx 24$ vs. 1) $\Rightarrow$ the effect is **not solely an SA artifact**.
- **Precision floor.** Peak couplers map to Ising $J \approx 0.03$, near the scale where integrated control errors (ICE) matter; shrinking the problem scale drops them below ICE and **erases the peak**.

10 Conclusion

Failure-conditioned ΔE is the difficulty diagnostic and **HD** its complement — the all-sample mean (hence the success rate) hides the peak, falling at large α only because more reads succeed while the remaining failures stay deep. Whenever an α -like hardness parameter is swept, we recommend reporting **failure-conditioned ΔE (floor & peak) together with HD over a fixed set of hardest instances**. Because the curve appears on *both* SA and a quantum annealer, it characterizes **structure of the benchmark instances**, though absolute values remain solver- and schedule-dependent.